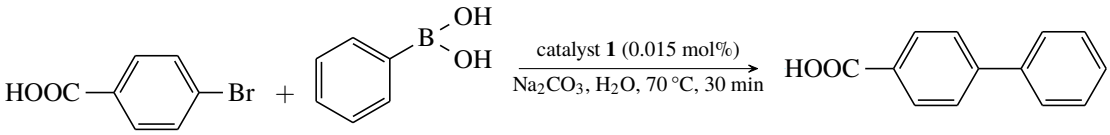
ORGANIC EXPERIMENT 1 — PRE-LAB CHECKLIST

Suzuki–Miyaura Cross-Coupling in Water: 4-Phenylbenzoic Acid

Condensed from the CM3191 Organic Lab Manual AY26/27 (Dr Hoang T. G., pp. 2–7) and lecture CM3191 Org L1. Week 7, **Thursday 1 October 2026**, Thursday lab (Dr Hoang T. G.). Venue **[CONFIRM: not stated in the organic schedule]**.



1. Reagent table (manual values; blanks in the manual filled by calculation)

Reagent	MW / conc.	Amount	mmol	Equiv.	Role
4-Bromobenzoic acid	201.02	0.34 g	1.69	1.00	aryl halide (limiting)
Phenylboronic acid	121.93	0.25 g	2.05	1.21	organoboron partner
Na_2CO_3 solution	0.5 M	10 mL	5.0	2.96	base
Pd catalyst solution **1**	0.25 mM (Pd)	1.0 mL	0.00025	0.000148	catalyst (0.0148 mol%)
4-Phenylbenzoic acid	198.22	**0.335 g (theor.)**	1.69	—	product

Theoretical yield = $n(\text{ArBr}) \times 198.22$. Recalculate with your *actual* mass of 4-bromobenzoic acid: $m_{\text{theor}} = m_{\text{ArBr}} \times (198.22/201.02) = 0.9861\, m_{\text{ArBr}}$.

2. Bring / prepare before the session

☐ Official datasheet (manual p. 7), **printed**
☐ Lab coat, safety goggles, nitrile gloves, covered shoes
☐ Labelled plastic bag for product (name)
☐ Know which hotplate is **shared (70 °C)** vs. your own
☐ This bench sheet (pp. 2–5)
☐ Phone for photos of every balance reading
☐ Pencil + marker for NMR tube label
☐ NMR solvent: DMSO-d_6 **[CONFIRM: manual text only says "the NMR solvent"]**

3. Session timeline (estimate)

#	Stage	Approx. time
1	Weigh solids, dissolve in Na_2CO_3 (own hotplate, vigorous stirring)	15 min
2	Add Pd solution, react in shared 70 °C block	**30 min** (fixed)
3	Cool, acidify with 1 M HCl in ice bath, stir 10 min, suction filter	25 min
4	Recrystallize from EtOH (heat, cool to RT 10–15 min, ice 15 min)	40 min
5	Suction filter, dry under IR lamp 15 min	20 min
6	Weigh, IR, NMR sample, datasheet photo check and GTA signature	20 min

SAFETY — from the manual (p. 3)

4-Bromobenzoic acid, phenylboronic acid, Na_2CO_3, the ligand and $Pd(OAc)_2$ may be harmful if inhaled, swallowed or absorbed through the skin; $Pd(OAc)_2$ and Na_2CO_3 are irritants; the catalyst solution should be considered harmful. NaOH is caustic; HCl is corrosive. Ethanol is highly flammable and a skin/eye irritant. DMSO-d_6 is absorbed through the skin and can carry dissolved substances with it. The product is not considered harmful, but avoid inhalation and skin contact. Perform all operations in a well-ventilated fume cupboard. **Added notes (not in the manual).** Ethanol is heated to gentle boiling in the recrystallization: hotplate only, no open flames, never heat a closed vessel. Acidification releases CO_2 (the manual's "gas will form"): add HCl dropwise, in the open conical flask. Waste: Pd-containing aqueous filtrate → metal aqueous waste; ethanol mother liquor → non-halogenated organic waste. **[CONFIRM: waste bins in the organic lab]**

BENCH MANUAL (ENGLISH)

Suzuki–Miyaura Cross-Coupling in Water

Tick each box as you complete the step. "Why" notes explain the purpose; they are not extra steps.

3.1 Reaction setup

☐ 1. Switch on the **shared** hotplate stirrer with heating block, set **70 °C**. *It needs time to equilibrate — do this first.*

☐ 2. Weigh **0.34 g 4-bromobenzoic acid** and **0.25 g phenylboronic acid** into a **20 mL reaction vial**; add a stir bar. Photograph both readings.

☐ 3. Add **10 mL 0.5 M Na_2CO_3**, cap, stir **vigorously on the hotplate stirrer in your own fume cupboard** (not the shared 70 °C block) until **all solids dissolve**.
Why: the base converts the acid into its water-soluble sodium carboxylate and activates the boronic acid. The catalyst must not be added until the solution is homogeneous.

☐ 4. Add **1.0 mL Pd catalyst solution (0.25 mM)**, cap, place in the preheated **70 °C** block. **Start the timer: 30 min.** Record the time.

3.2 Work-up

☐ 5. Remove the vial, cool to room temperature, transfer to a **250 mL conical flask**; stir in an **ice bath**.

☐ 6. Add **20 mL 1 M HCl slowly, dropwise**, with stirring. **Gas (CO_2) forms**; a solid is expected to precipitate.
Why: HCl first destroys the carbonate, then protonates the carboxylate; the neutral acid is insoluble in cold water. 20 mmol HCl exceeds the 10 mmol needed for 5 mmol carbonate, so the mixture ends strongly acidic.

☐ 7. Check **pH paper: acidic**. Add **20 mL water**; stir **10 min** in the ice bath.

☐ 8. Collect the crude solid by **suction filtration** (Appendix A).

3.3 Recrystallization (single solvent: ethanol)

☐ 9. Crude solid + stir bar into a **50 mL conical flask** (or a 50 mL beaker). Add **3 mL 1 M HCl** and **10 mL ethanol** (= solution 1).

☐ 10. Second 50 mL conical flask: **20 mL ethanol** + a few boiling chips (= solution 2).

☐ 11. Heat both to **gentle boiling**. With a dropper, add hot ethanol from solution 2 to solution 1 **only until all solid dissolves** — use the **minimum**.
Why: every extra mL keeps product in the mother liquor and lowers the yield. The HCl keeps the product fully protonated (neutral acid, not its salt).

☐ 12. Cool **undisturbed to RT (10–15 min)**, then place **both** flasks in ice for **15 min**. *Solution 2 becomes your cold wash solvent.*

☐ 13. Suction filter; if necessary, transfer and wash with a **minimum of cold ethanol** (solution 2).

☐ 14. Crystals onto a large filter paper labelled with your name; dry under the **IR lamp, 15 min**.

3.4 Characterization

☐ 15. Weigh the product in a **pre-weighed plastic bag**; photograph; calculate % yield.

☐ 16. Record an **IR spectrum**; note the file name.

☐ 17. NMR: **∼10 mg** in a clean, dry plastic vial, dissolve in **0.6 mL NMR solvent**, transfer with a clean, dry glass dropper; solution height **∼5 cm** (top up with the same solvent). Cap and label.

☐ 18. Show photos to the GTA; get the **printed official datasheet signed**.

실험 매뉴얼 (한국어)

물에서의 Suzuki–Miyaura 교차 커플링

단계를 마칠 때마다 체크하세요. “이유”는 목적 설명이며 추가 단계가 아닙니다.

3.1 반응 준비

☐ 1. **공용** 핫플레이트(히팅 블록)를 켜고 **70 °C**로 설정합니다. 온도가 안정되는 데 시간이 걸리므로 가장 먼저 합니다.

☐ 2. **20 mL 반응 바이알**에 **4-bromobenzoic acid 0.34 g**, **phenylboronic acid 0.25 g**, 교반 막대를 넣습니다. 저울 값은 각각 사진으로 남깁니다.

☐ 3. **0.5 M Na_2CO_3 10 mL**를 넣고 뚜껑을 닫은 뒤, **본인 후드의 교반기**(공용 70 °C 블록 아님)에서 **고체가 모두 녹을 때까지** 세게 교반합니다.
이유: 염기가 산을 수용성 나트륨 카복실산염으로 바꾸고 보론산을 활성화합니다. 균일한 용액이 되기 전에는 촉매를 넣지 않습니다.

☐ 4. **Pd 촉매 용액(0.25 mM) 1.0 mL**를 넣고 뚜껑을 닫아 예열된 **70 °C** 블록에 올립니다. **30분 타이머 시작**, 시각을 기록합니다.

3.2 후처리

☐ 5. 바이알을 내려 실온으로 식힌 뒤 **250 mL 삼각플라스크**로 옮기고 **얼음 수조**에서 교반합니다.

☐ 6. **1 M HCl 20 mL를 한 방울씩 천천히** 넣습니다. **기체(CO_2)가 발생**하고, 고체가 석출될 것으로 예상됩니다.
이유: HCl은 먼저 탄산염을 분해하고, 이어 카복실산염을 양성자화합니다. 중성 산은 찬물에 녹지 않습니다. 탄산염 5 mmol을 분해하는 데 필요한 산은 10 mmol인데 HCl은 20 mmol이므로, 최종 용액은 강한 산성이 됩니다.

☐ 7. **pH 시험지로 산성** 확인 후 **물 20 mL**를 넣고 얼음 수조에서 **10분** 교반합니다.

☐ 8. **감압 여과**로 조생성물을 얻습니다(부록 A).

3.3 재결정 (단일 용매: 에탄올)

☐ 9. 조생성물과 교반 막대를 **50 mL 삼각플라스크**(또는 50 mL 비커)에 넣고 **1 M HCl 3 mL**, **에탄올 10 mL**를 더합니다(용액 1).

☐ 10. 다른 50 mL 삼각플라스크에 **에탄올 20 mL**와 끓임쪽 몇 개를 넣습니다(용액 2).

☐ 11. 두 용액을 **약하게 끓을 때까지** 가열합니다. 스포이트로 용액 2의 뜨거운 에탄올을 용액 1에 **고체가 다 녹을 만큼만** 넣습니다. **최소량**이 핵심입니다.
이유: 에탄올을 더 넣을수록 모액에 남는 생성물이 늘어 수율이 떨어집니다. HCl은 생성물을 염이 아닌 중성 산 상태로 유지합니다.

☐ 12. **흔들지 말고 실온까지 10–15분** 식힌 뒤, **두 플라스크 모두** 얼음에 **15분** 둡니다. 용액 2는 차가운 세척 용매로 씁니다.

☐ 13. 감압 여과합니다. 옮기거나 세척할 필요가 있으면 **차가운 에탄올(용액 2)을 최소량**만 씁니다.

☐ 14. 이름을 적은 큰 거름종이에 결정을 펴서 **IR 램프 아래 15분** 건조합니다.

3.4 분석

☐ 15. **미리 무게를 잰 비닐봉투**에 담아 무게를 재고 사진을 찍은 뒤 수율을 계산합니다.

☐ 16. **IR 스펙트럼**을 측정하고 파일명을 적어 둡니다.

☐ 17. NMR: 깨끗하고 마른 플라스틱 바이알에 **약 10 mg**, **NMR 용매 0.6 mL**로 녹이고 깨끗하고 마른 유리 스포이트로 옮깁니다. 용액 높이 **약 5 cm**(부족하면 같은 용매로 보충), 뚜껑과 라벨.

☐ 18. 사진을 GTA에게 보여 주고 **인쇄한 공식 datasheet에 서명**을 받습니다.

DATASHEET — WORKING LOG

Suzuki–Miyaura Cross-Coupling in Water

Name: KIM DONGHWAN Fumehood: _____ Date: 1 Oct 2026 Photos taken: □

1. Masses

4-Bromobenzoic acid (g)		Phenylboronic acid (g)	
n(ArBr) = mass/201.02 (mmol)		Equiv. $PhB(OH)_2$	
Empty plastic bag (g)		Bag + product (g)	
Product (g)		Theoretical (g) = $0.9861\, m_{\text{ArBr}}$	
		Percentage yield (%)	

2. Time log

Pd added (clock)	Out of block	HCl addition done	Into IR lamp

3. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Solids + Na_2CO_3, stirred	all reactants dissolve [M]; clear, colourless [E]		
Pd catalyst solution	pale yellow [M, App. 1]		
After 30 min at 70 °C	*record colour and any solid or grey tint (Pd black)* [E]		
HCl addition	gas forms [M] (CO_2 [C]); solid precipitates [E]		
pH paper after HCl	acidic [M]; calc. pH ≈ 0.7 [C]		
Hot ethanol, solution 1	all material dissolves with minimal ethanol [M]		
Cooling to RT, then ice	crystals form (recrystallisation) [M]		
Dried product	white solid [E]		

Sources: [M] stated in the lab manual; [C] calculated from manual quantities; [E] general expectation, not stated in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see; a mismatch is a result worth discussing.

4. Ethanol used in recrystallization & catalyst metrics

Hot EtOH added from sol. 2 (mL, est.)		n(product) (mmol)	
TON = $n_{\text{product}}/(2.5 \times 10^{-4}$ mmol)		TOF = TON/30 min (1/min)	
IR file name		NMR tube no. / solvent	

5. Other observations

To be completed at the end of the session		
Student signature	GTA name	GTA signature

This is a working log. The **official** datasheet (lab manual p. 7) must still be printed, signed by the GTA and submitted with the report.

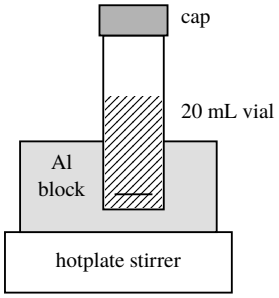
APPENDIX A — APPARATUS

Reaction, acidification, suction filtration and two-flask recrystallization

Schematic, not to scale.

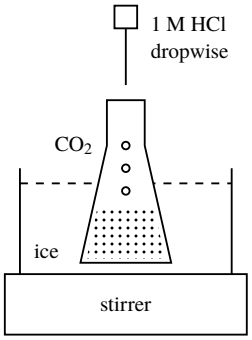
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(1) Reaction



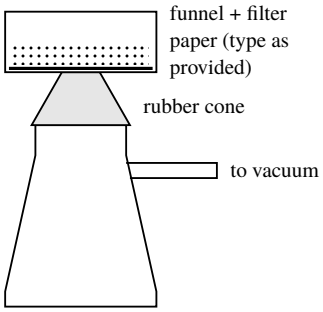
70 °C, 30 min

(2) Acidification



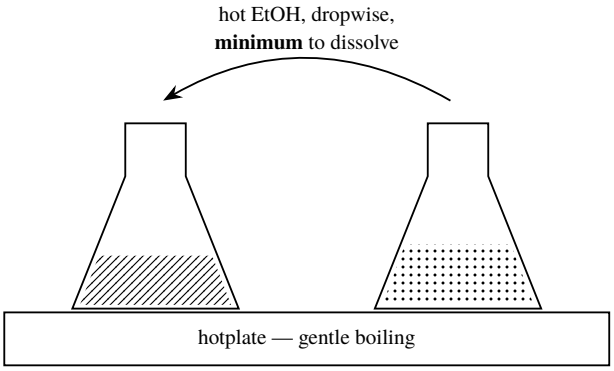
250 mL conical, stirred

(3) Suction filtration



filter flask

(4) Two-flask recrystallization



Solution 1: crude + 3 mL 1 M HCl + 10 mL EtOH + stir bar

Solution 2: 20 mL EtOH + boiling chips

ORDER OF EVENTS AFTER DISSOLUTION

Clear solution → remove from heat → **undisturbed** to RT (10–15 min, larger and purer crystals) → ice 15 min (**solution 2 too**, as cold wash) → suction filter → wash with minimum cold EtOH → IR lamp 15 min.

Tip (not in the manual). If too much ethanol was added and few crystals form, gently boil off some ethanol on the hotplate and cool again; record this on the datasheet.